AKSHAY REDDY VELUGATI

Warangal, Telangana

+91 8790756797 | akshayvelugati@gmail.com | akshayvelugati | akshayreddyvelugati

Summary

Final-year B.Tech (AI & Data Science) undergraduate with experience in Python, SQL, data analytics and machine learning. Skilled in building analytics-driven applications, dashboards, and predictive models, with hands-on experience in educational analytics and AI-enabled systems.

Education

Amrita Vishwa Vidhyapeetham **2023 – 2027**

Bachelor of Technology – Artificial Intelligence and Data Science **CGPA: 7.72/10** Amritapuri, India

Resonance Junior College **2021 - 2023**

Higher Secondary Education(TSBIE) - PCM **Percentage: 92.8 %** Warangal, India

Experience

Research Intern – Biomedical Learning Analytics **Jan 2026 – Apr 2026**

Amrita Mind Brain Center (MoE Virtual Labs Initiative)

- Developed virtual biomedical laboratory modules covering EMG, ECG, HRV, and pulse oximetry for simulation-based learning.
- Built learner analytics workflows and dashboards to analyze engagement, assessment performance, and learning outcomes.
- Collaborated with researchers and developers to integrate simulation-based learning, analytics, and scalable application workflows.

Projects

NeuroGuideLab – Neuro-Symbolic Learning Platform **Neuro-Symbolic AI, XAI (2026)**

- Developed a web-based learning platform integrating biomedical simulations, assessments, and learner analytics.
- Designed assessment workflows to capture learner interactions, pre/post-test performance, checkpoint completion, and engagement metrics.
- Built analytical dashboards to visualize learning gains, engagement trends, and instructional outcomes.
- Analyzed pilot-study data from approximately 40 learners to evaluate platform usability, engagement, and learning outcomes.

Skysail – Air Travel Management System **React.js, Node.js, PostgreSQL, (2025)**

- Developed a full-stack airline management platform using React.js and Node.js, supporting passenger management, flight scheduling, and airport operations.
- Designed normalized PostgreSQL database schemas and REST APIs for efficient data storage, retrieval, and application workflows.
- Implemented backend workflows and optimized SQL queries to improve application performance and reliability.

Publications

Weather-Aware Solar Energy Forecasting Using a Hybrid ARIMA–LSTM Model (Proceedings of IC³ 2025)

- Developed a hybrid forecasting model combining ARIMA and LSTM with preprocessing and feature engineering using NASA POWER weather datasets (2020–2025).
- Evaluated forecasting performance using MAE, RMSE, and MAPE to compare hybrid and standalone forecasting models.
- Achieved a 42% reduction in Mean Absolute Percentage Error (MAPE) compared to standalone forecasting approaches.

Skills

- **Programming:** Python, SQL, Java, JavaScript
- **Data & Analytics:** Pandas, NumPy, Matplotlib, Tableau
- **AI & Machine Learning:** Generative-AI,Prompt engineering, RAG, Scikit-learn, TensorFlow, PyTorch, Keras
- **Databases:** PostgreSQL, Supabase, BigQuery, Database Design
- **Web & Backend:** React.js, Node.js, REST API, HTML5, CSS3, Tailwind CSS, Vite
- **Developer Tools:** Git, GitHub, VS Code
- **Soft Skills:** Communication, Presentation, Leadership, Team Collaboration, Analytical Thinking

Certifications

- Elements of Ai - Minnalearn, University of Helsinki (2025)
- Foundations of Deep learning - NVIDIA DLI. (2025)

Achievements

- Event Anchor(MC) - Hosted multiple college events
- National-Level Swimmer